

Wasserstein–Fisher–Rao Gaussian Gradient Flow: Phase Separation Between Transport and Natural-Gradient Descent

Abstract

We study the Wasserstein–Fisher–Rao (WFR) gradient flow on the Gaussian variational manifold, a hybrid of the Bures–Wasserstein transport flow and the Fisher–Rao (natural-gradient) flow with a tunable transport strength λ_t . We discretize the flow by forward–backward operator splitting – one Wasserstein half-step followed by one Fisher–Rao half-step per iteration – and compare it against the two pure flows ($\lambda \equiv 0$, Fisher–Rao only; and Wasserstein only) on two ill-conditioned two-dimensional targets: an exact anisotropic Gaussian posterior and a smooth nonseparable strongly log-concave posterior. From a far, underdispersed initialization the experiments expose a clean *phase separation*: the Wasserstein transport drives fast early descent (warmup) but stalls in the tail, while the Fisher–Rao step is slow during warmup but converges rapidly once the covariance is locally calibrated to the curvature. The WFR splitting inherits both phases, and a curvature-adaptive transport schedule $h_n = h_{\max}/(1 + (s_n/s_0)^2)$ – large transport when the state is underdispersed, decaying transport as the covariance calibrates – realizes the hand-off automatically.

Measured against the energy gap versus iteration, the verdict is decisive. The WFR splitting reaches every tolerance in roughly half the iterations of pure Fisher–Rao: the curvature-adaptive schedule reaches gap $< 10^{-3}$ in 15–22 iterations against Fisher–Rao’s 31–41, and dominates the fixed and theorem-bound schedules among WFR variants – the theorem bound $h = \mu_{\min}$ is far too conservative once the state is underdispersed. Wasserstein-only is decisively worst: it collapses the bulk of the initial gap in the first few iterations but then plateaus, taking thousands of iterations to reach 10^{-6} (and never reaching it on the Gaussian target at $\Lambda = 1000$) because transport alone inflates the covariance volume without ever calibrating its shape.

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1 Setup

Gaussian variational family and energy. We approximate a target posterior $\rho_{\text{post}} \propto e^{-V}$ on $\mathbb{R}^d$ ($d = 2$ throughout) by a Gaussian $\mathcal{N}(m, C)$, $C \in \text{SPD}(d)$, parameterized by the state $a = (m, C)$. The variational objective is the (reverse) KL up to an a -independent constant,

$$\mathcal{E}(a) = \mathbb{E}_{\theta \sim \mathcal{N}(m, C)}[V(\theta)] - \frac{1}{2} \log \det C, \quad \mathcal{E}(a) - \mathcal{E}(a_\star) = \text{KL}(\mathcal{N}(m, C) \parallel \rho_{\text{post}}), \quad (1)$$

minimized at $a_\star = (m_\star, C_\star)$. We write the posterior log-gradient and log-Hessian expectations

$$g = \mathbb{E}_{\mathcal{N}(m, C)}[\nabla \log \rho_{\text{post}}] = -\mathbb{E}[\nabla V], \quad H = \mathbb{E}_{\mathcal{N}(m, C)}[\nabla^2 \log \rho_{\text{post}}] = -\mathbb{E}[\nabla^2 V], \quad (2)$$

so that $-H = \mathbb{E}[\nabla^2 V] \succeq 0$ for a log-concave target. We assume global strong log-concavity and smoothness, $\alpha I \preceq \nabla^2 V \preceq \beta I$.

The WFR Gaussian flow. The Bures–Wasserstein gradient flow of $\mathcal{E}$ moves mass by transport; the Fisher–Rao flow rescales mass by the natural gradient. The WFR flow combines them with transport strength $\lambda_t \geq 0$:

$$\dot{m} = (C + \lambda_t I) g, \quad \dot{C} = \underbrace{C + CHC}_{\text{Fisher–Rao}} + \lambda_t \underbrace{(2I + CH + HC)}_{\text{Wasserstein}}. \quad (3)$$

Setting $\lambda_t \equiv 0$ recovers the Fisher–Rao (natural-gradient) flow; dropping the Fisher–Rao terms recovers the Bures–Wasserstein flow. The additive $2\lambda_t I$ term is the source of the warmup advantage analyzed below: it inflates an underdispersed covariance at a rate independent of the current scale.

Targets. The anisotropy parameter is Λ (the symbol λ is reserved for the transport strength λ_t). We use two targets.

- **Gaussian posterior:** $\rho_{\text{post}} = \mathcal{N}(0, \Gamma)$ with $\Gamma = \text{diag}(1, \Lambda)$ the *covariance*. Then $V(\theta) = \frac{1}{2} \theta^\top \Gamma^{-1} \theta$, so $g = -\Gamma^{-1} m$ and $H = -\Gamma^{-1}$ are exact and state-independent, $\alpha = 1/\Lambda$, $\beta = 1$, and the optimum is closed form $m_\star = 0$, $C_\star = \Gamma$ with an analytic energy gap (1).
- **Smooth log-cosh posterior:** the nonseparable strongly log-concave

$$V(x, y) = \frac{1}{20} (\sqrt{\Lambda} x - y)^2 + \frac{\delta}{2} (x^2 + y^2) + \gamma \log \cosh(y), \quad \delta = 0.05, \gamma = 1, \quad (4)$$

with $\alpha = \delta$ and the eigenvalue smoothness constant $\beta = \lambda_{\max} \begin{pmatrix} \delta + \Lambda/10 & -\sqrt{\Lambda}/10 \\ -\sqrt{\Lambda}/10 & \delta + 1/10 + \gamma \end{pmatrix}$. Expectations of $\tanh$, sech^2 and $\log \cosh$ under the marginal $Y \sim \mathcal{N}(m_y, C_{22})$ are evaluated by deterministic Gauss–Hermite quadrature; the optimum is found numerically.

The target metadata, including β/α (the condition number that governs the tail rate), is reported in Table 1.

Initialization. Every run starts far and underdispersed: $C_0 = \varepsilon I$ and the mean placed a distance $r = 8$ along the flat-valley direction $(1, \sqrt{\Lambda})/\sqrt{1 + \Lambda}$. This is the regime that separates the two flows – the covariance must inflate by orders of magnitude and the mean must traverse the valley.

2 Forward–backward discretization and the five methods

Operator splitting. We discretize (3) by forward–backward operator splitting with Fisher–Rao step Δt and Wasserstein step $h_n = \lambda_n \Delta t$. Each iteration applies one Wasserstein half-step then one Fisher–Rao half-step. With g_n, H_n from (2) evaluated at $a_n = (m_n, C_n)$, the *Wasserstein half-step* is

$$\begin{aligned} m_{n+1/2} &= m_n + h_n g_n, & M_n &= I + h_n H_n, & \tilde{C} &= M_n C_n M_n, \\ C_{n+1/2} &= \frac{1}{2} \left(\tilde{C} + 2h_n I + [\tilde{C}(\tilde{C} + 4h_n I)]^{1/2} \right), \end{aligned} \quad (5)$$

and, with $g_{n+1/2}, H_{n+1/2}$ re-evaluated at $a_{n+1/2}$, the *Fisher–Rao half-step* is the KL/natural-gradient update

$$m_{n+1} = m_{n+1/2} + \Delta t C_{n+1/2} g_{n+1/2}, \quad C_{n+1} = (1 + \Delta t)(C_{n+1/2}^{-1} - \Delta t H_{n+1/2})^{-1}. \quad (6)$$

The matrix square root in (5) is computed spectrally: $\tilde{C}$ and $\tilde{C} + 4h_n I$ commute, so with the symmetric eigendecomposition $\tilde{C} = V \text{diag}(w) V^\top$,

$$C_{n+1/2} = V \text{diag}\left(\frac{1}{2}(w + 2h_n + \sqrt{w(w + 4h_n)})\right) V^\top, \quad (7)$$

which is exact, symmetric, and SPD whenever $\tilde{C} \succeq 0$ and $h_n \geq 0$. The Fisher–Rao covariance update (6) is unconditionally SPD for log-concave targets ($-H \succeq 0$). At $h_n = 0$ the Wasserstein half-step is the identity and the iteration reduces exactly to the pure Fisher–Rao step.

The five methods.

- **FR-only** ($\lambda \equiv 0$): pure Fisher–Rao, $h_n = 0$.
- **W-only**: the Wasserstein half-step (5) alone with the fixed $h_n = c/\beta$.
- **WFR-fixed**: the full splitting with fixed $h_n = c/\beta$.
- **WFR-theory**: the full splitting with the constant $h_n = \mu_{\min} = \min(\lambda_{\min}(C_0), 1/\beta)$, the maximizer of the proven discrete transport contribution $f(h) = \alpha h / (1 + h/\mu_{\min})^2$.
- **WFR-adaptive**: the full splitting with the curvature-adaptive schedule

$$h_n = \frac{h_{\max}}{1 + (s_n/s_0)^2}, \quad h_{\max} = \frac{0.9}{\beta}, \quad s_0 = 0.5, \quad s_n = \lambda_{\min}(C_n^{1/2}(-H_n)C_n^{1/2}). \quad (8)$$

The calibration ratio s_n is the smallest eigenvalue of the dimensionless curvature-in-covariance-metric operator: $s_n \ll 1$ signals an underdispersed state (the Fisher–Rao step is throttled, large transport helps warmup), while $s_n \approx 1$ signals a locally calibrated covariance (the Fisher–Rao step is reliable, transport should decay). For the Gaussian target at $C = \Gamma$, $-H = \Gamma^{-1}$ gives $C^{1/2}(-H)C^{1/2} = I$ and $s_n = 1$ exactly.

3 Results

Unless stated otherwise the head-to-head uses a shared Fisher–Rao step $\Delta t = 0.5$ and the fixed fraction $c = 0.5$ for W-only and WFR-fixed, on the production grid $\Lambda \in \{100, 1000\}$, $\varepsilon \in \{10^{-2}, 10^{-3}\}$. Every run is stepped to a fixed iteration budget. Target constants are in Table 1.

Table 1: Target metadata: strong-log-concavity α , smoothness β , condition number β/α , initial covariance scale ε , and the initial energy gap $\text{gap}_0 = \mathcal{E}(a_0) - \mathcal{E}(a_\star)$.

target	Λ	ε	α	β	β/α	gap_0
Gaussian	100	0.001	0.010	1.000	100.0	8.8
Gaussian	100	0.01	0.010	1.000	100.0	6.5
Gaussian	1000	0.001	0.001	1.000	1000.0	9.4
Gaussian	1000	0.01	0.001	1.000	1000.0	7.1
smooth log-cosh	100	0.001	0.050	10.161	203.2	14
smooth log-cosh	100	0.01	0.050	10.161	203.2	12
smooth log-cosh	1000	0.001	0.050	100.151	2003.0	13
smooth log-cosh	1000	0.01	0.050	100.151	2003.0	11

3.1 Phase separation: transport wins warmup, Fisher–Rao wins the tail

Figures 1–2 show the energy gap against iteration, with a warmup zoom (left) and the full trajectory (right). Three facts are consistent across every (Λ, ε) :

1. **W-only descends fastest in the first few iterations** but *plateaus*: from $C_0 = \varepsilon I$ the additive $2h_n I$ term in (5) inflates the covariance at a scale-independent rate, collapsing the bulk of the initial gap quickly, after which the lack of the natural-gradient rescaling leaves it stalled far from $a_\star$.
2. **FR-only is slow during warmup** – the natural-gradient covariance inflates only geometrically and the mean update $\Delta t Cg$ is throttled by the tiny C – **but converges fastest among the pure flows in the tail** once the covariance is calibrated.
3. **WFR (fixed and adaptive) inherit both phases**: they track W-only’s fast warmup and then FR-only’s fast tail. This makes them the fastest methods overall – the adaptive schedule reaches $\text{gap} < 10^{-3}$ in 15–22 iterations, about half of FR-only’s 31–41 (Table 2), and reaches 10^{-6} in 23–35 iterations across the whole grid.

3.2 The mechanism: the covariance, not the mean, is the bottleneck

The phase separation is not just a property of the scalar energy gap – it has a concrete mechanism, visible once the gap is decomposed into its mean and covariance parts. Figure 3 plots $\|m_n - m_\star\|$ and $\|C_n - C_\star\|_F$ separately on the hardest start ($\Lambda = 1000$, $\varepsilon = 10^{-3}$). Three observations pin down why the flows behave as they do:

1. **The covariance dominates the error budget.** On the Gaussian target the covariance error at the underdispersed start is $\approx 10^3$ (it must inflate from εI to a top eigenvalue $\Lambda = 10^3$) while the mean error is only ≈ 8 . The optimization is, to leading order, a covariance-inflation problem; the mean is a sideshow.
2. **The mean always calibrates before the covariance.** For every convergent method the mean error crosses 10^{-3} tens of iterations before the covariance error does (e.g. on the smooth target, FR-only at 39 vs 52 iterations; WFR-adaptive at 24 vs 42). The tail rate is therefore set entirely by how fast the covariance is calibrated – which is exactly the quantity the Fisher–Rao step accelerates and the adaptive ratio s_n measures.

Table 2: Iterations to reach $\text{gap} < 10^{-3}$ at the harder start $\varepsilon = 10^{-3}$. WFR-adaptive needs the fewest iterations – about half of FR-only – and W-only is an order of magnitude behind.

target	method	iterations to $\text{gap} < 10^{-3}$	
		$\Lambda=100$	$\Lambda=1000$
Gaussian	W-only	575	3653
	FR-only	36	41
	WFR-fixed	16	22
	WFR-theory	31	37
	WFR-adaptive	15	21
smooth log-cosh	W-only	260	2560
	FR-only	31	31
	WFR-fixed	17	23
	WFR-theory	26	26
	WFR-adaptive	16	22

3. **W-only inflates volume but cannot calibrate shape.** On the Gaussian target its covariance error barely moves (it drifts from $\approx 10^3$ down to only $\approx 9.5 \times 10^2$ over the first 60 iterations) because the additive $2h_n I$ term grows the volume isotropically without ever matching the target’s anisotropy. This is the mechanism behind its plateau: transport produces a big but mis-shaped covariance, and only the natural-gradient rescaling can fix the shape.

3.3 The adaptive schedule tracks curvature calibration

Figure 4 shows the normalized transport $h_n \beta$ and the calibration ratio s_n along the WFR-adaptive runs. The schedule holds transport near the ceiling $h_{\max} \beta = 0.9$ while $s_n \ll s_0$ (the underdispersed warmup) and decays it as s_n approaches 1 (the covariance becomes locally calibrated and the Fisher–Rao step takes over), exactly the hand-off implied by the phase separation.

3.4 Iterations to convergence

Table 2 reports, for the harder underdispersed start $\varepsilon = 10^{-3}$, the iterations each method needs to reach $\text{gap} < 10^{-3}$, and Table 3 gives the full per-run breakdown across thresholds 10^{-1} , 10^{-3} , 10^{-6} .

- **The WFR splitting wins outright.** WFR-adaptive reaches the tolerance in 15–22 iterations against FR-only’s 31–41 – roughly half, because each iteration blends a transport move and a natural-gradient move – and WFR-fixed is close behind. The advantage is largest at the harder, more underdispersed starts, where the transport warmup matters most.
- **W-only is decisively worst.** It collapses the bulk of the gap in the first handful of iterations but then plateaus far from the optimum, taking 575–3653 iterations to reach 10^{-3} and never reaching 10^{-6} on the Gaussian target at $\Lambda = 1000$ within the budget (Table 3); transport alone cannot calibrate the covariance.

All twenty production runs per method are SPD-feasible and monotone throughout.

3.5 How much transport? The schedule sweep

Figure 5 sweeps the fixed fraction c (so $h = c/\beta$, with $c = 0$ recovering FR-only); Table 4 gives the numbers. Increasing c *monotonically reduces the iteration count* until it saturates: more transport means each iteration makes more progress through the warmup. The theorem-bound schedule $h = \mu_{\min}$ (star markers) lands in the interior of the fixed- c curve rather than at an optimum: with the tiny $C_0 = \varepsilon I$ the bound gives $\mu_{\min} = \varepsilon$ on the Gaussian target (since $\beta = 1$), an extremely small transport that throttles the warmup the bound was meant to accelerate – direct evidence that the conservative proof-bound schedule underuses transport precisely in the underdispersed regime it targets.

3.6 Stability across the discretization step

The appendix heatmap (Figure 6) sweeps the Fisher–Rao step Δt against the transport fraction c for WFR-fixed, confirming that the splitting remains SPD-feasible and convergent across a broad joint range rather than at a single tuned Δt .

4 Findings and implications

1. **The two flows phase-separate.** Wasserstein transport, through its scale-independent $2h_n I$ covariance inflation, dominates the warmup; Fisher–Rao natural-gradient descent dominates the tail once the covariance is calibrated to the curvature. The separation is clean and holds across both targets and all (Λ, ε) .
2. **The WFR splitting combines the phases, halving iterations.** It reaches every tolerance in roughly half the iterations of pure Fisher–Rao (Table 2): WFR-adaptive hits gap $< 10^{-3}$ in 15–22 iterations against FR-only’s 31–41, and reaches 10^{-6} in 23–35. This is the headline result – combining transport and natural-gradient descent in one iteration is strictly faster than either pure flow.
3. **Adaptivity beats both fixed schedules, and the proof bound is too conservative.** Among WFR variants the curvature-adaptive schedule dominates fixed c and the theorem bound $h = \mu_{\min}$; the latter collapses to a negligible transport in exactly the underdispersed regime it was meant to accelerate.
4. **Transport alone is not enough.** W-only plateaus and never reaches 10^{-6} at $\Lambda = 1000$ on the Gaussian target: without the natural-gradient rescaling it cannot calibrate the covariance.

5 Limitations

The study is deliberately controlled and its scope bounds the claims. *(i) Dimension and targets:* $d = 2$, two globally strongly-log-concave targets; the per-iteration speedup constants will differ in higher dimension and the phase separation is not established for multimodal or heavy-tailed targets. *(ii) Exact expectations:* (g, H) are exact (closed form or Gauss–Hermite), so the comparison isolates the discretization, not Monte-Carlo gradient noise; under sampling noise the relative ranking could shift. *(iii) First-order splitting:* we use one Wasserstein then one Fisher–Rao half-step (Lie–Trotter); a Strang splitting or a single fused step could change the per-iteration progress and is untested. *(iv) The adaptive schedule* is one heuristic with hand-set $h_{\max} = 0.9/\beta$, $s_0 = 0.5$; we do not claim it is optimal, only that it dominates the fixed and proof-bound schedules tested.

A Discretization-step robustness

Gaussian posterior: WFR phase separation (gap vs iteration)

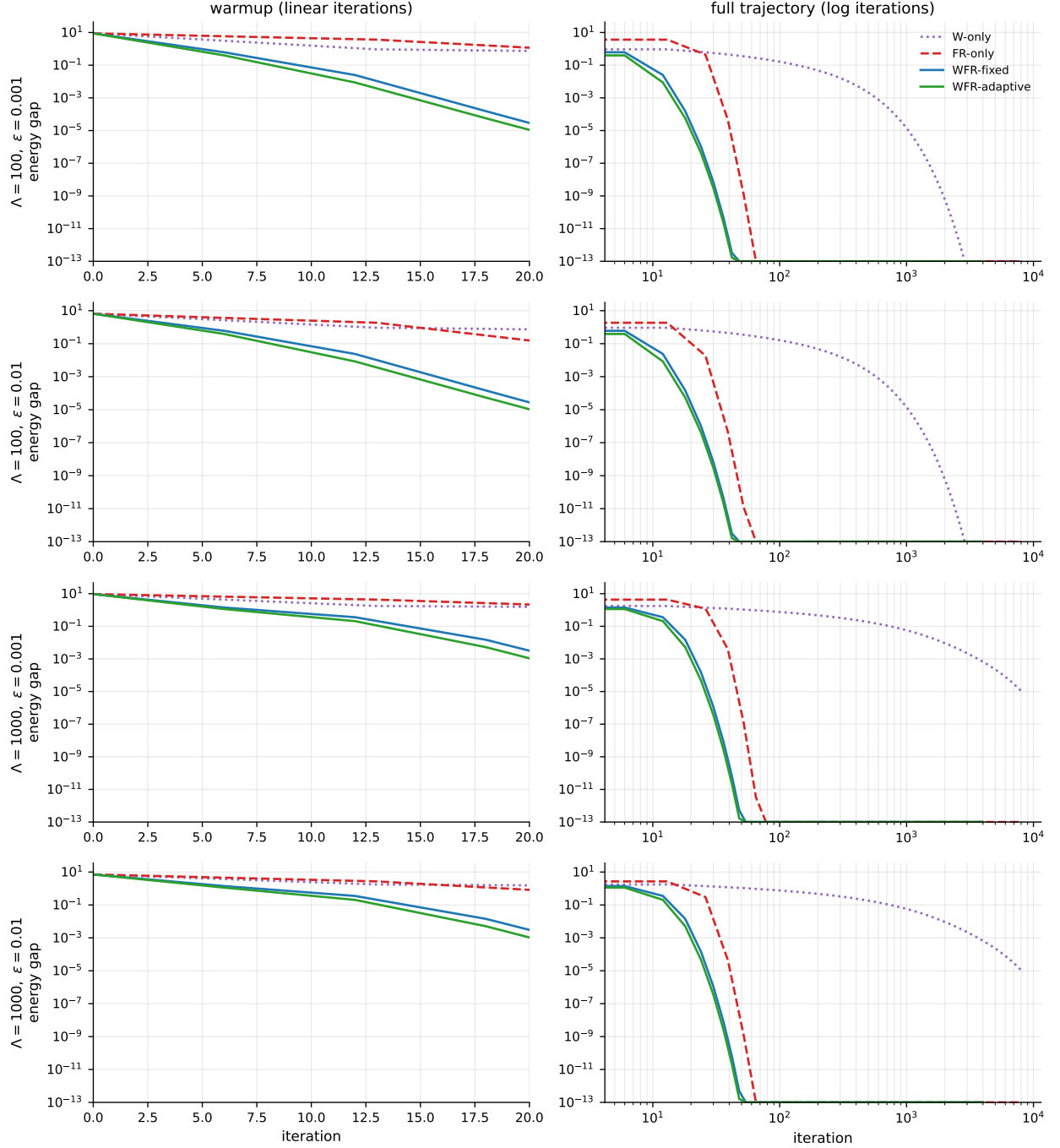


Figure 1: Gaussian posterior: energy gap vs iteration, warmup zoom (left) and full trajectory (right), one row per (Λ, ϵ) . W-only wins the first iterations then plateaus; FR-only lags then overtakes; WFR-adaptive captures both phases.

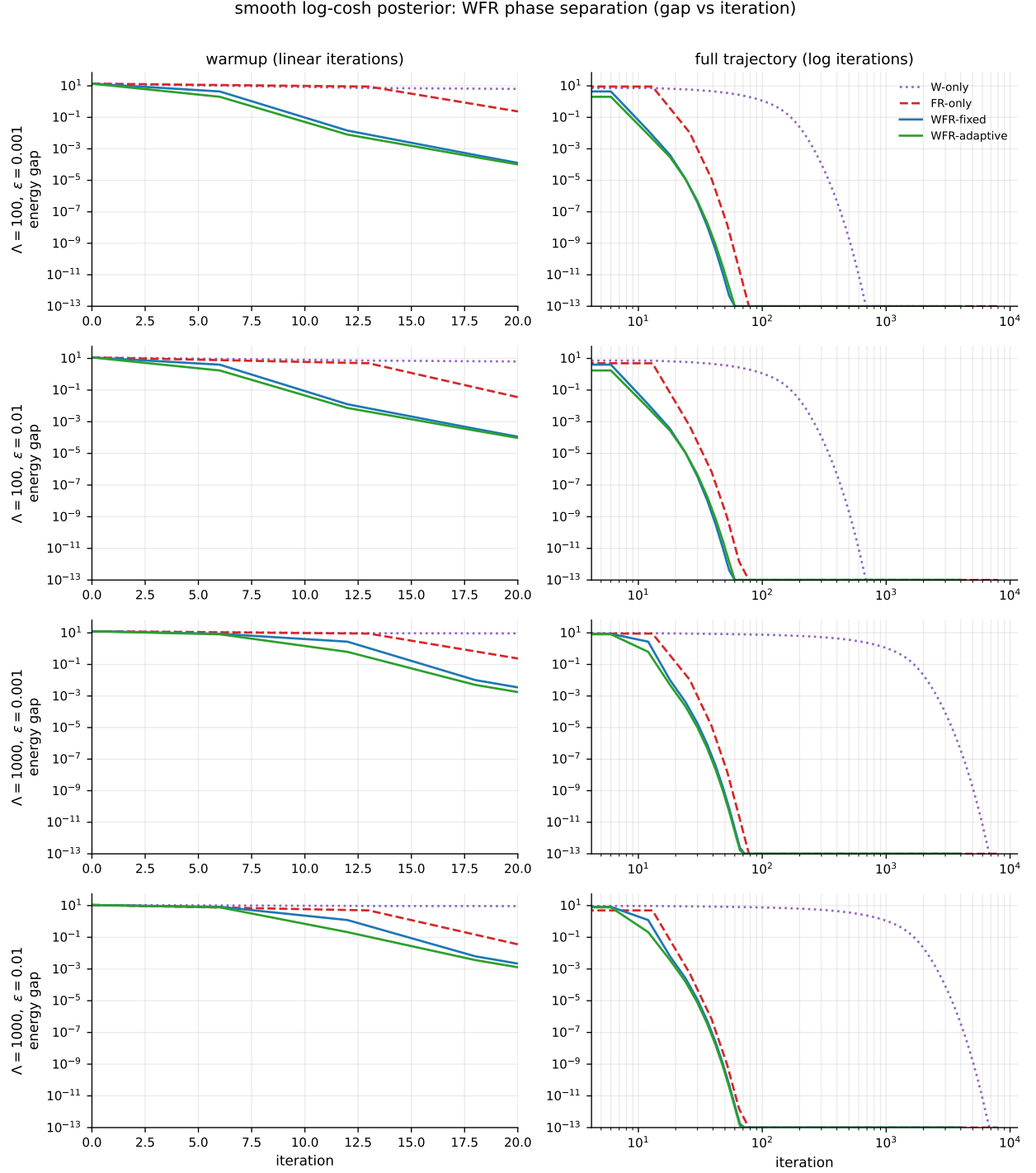


Figure 2: Smooth log-cosh posterior: same layout as Figure 1. The phase separation persists on the nonseparable, numerically-optimized target.

Phase-separation mechanism: the covariance, not the mean, is the bottleneck

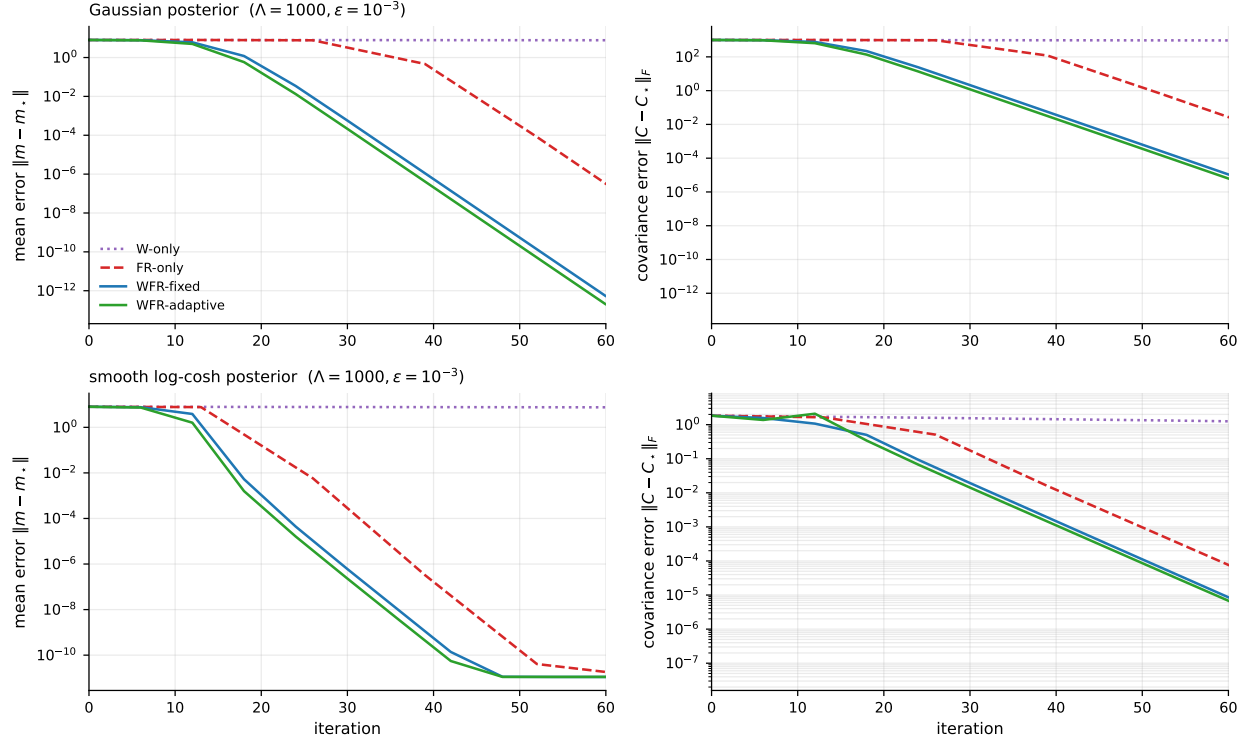


Figure 3: Phase-separation mechanism on the hardest start ($\Lambda = 1000, \varepsilon = 10^{-3}$): mean error $\|m - m_*\|$ (left) and covariance error $\|C - C_*\|_F$ (right), Gaussian (top) and smooth log-cosh (bottom), vs iteration. On the Gaussian target the covariance error starts two orders of magnitude above the mean error; W-only never collapses it, while FR-only and WFR do once the mean has arrived.

WFR-adaptive transport schedule and curvature calibration

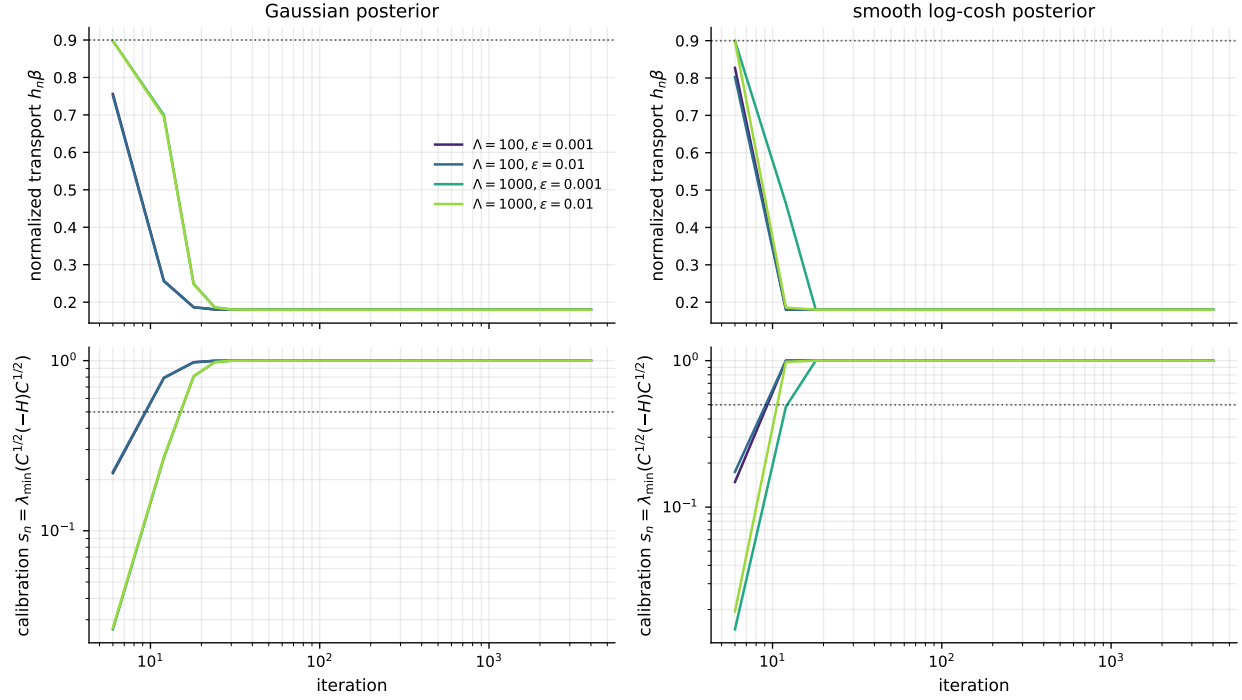


Figure 4: WFR-adaptive transport schedule. Top: normalized transport $h_n\beta$ (ceiling 0.9 dotted). Bottom: calibration ratio $s_n = \lambda_{\min}(C_n^{1/2}(-H_n)C_n^{1/2})$ (log scale, $s_0 = 0.5$ dotted), both vs iteration. Transport stays high through warmup and decays as the state calibrates.

WFR schedule sweep: transport cuts iterations to convergence; $\star$ = theory $h = \mu_{\min}$

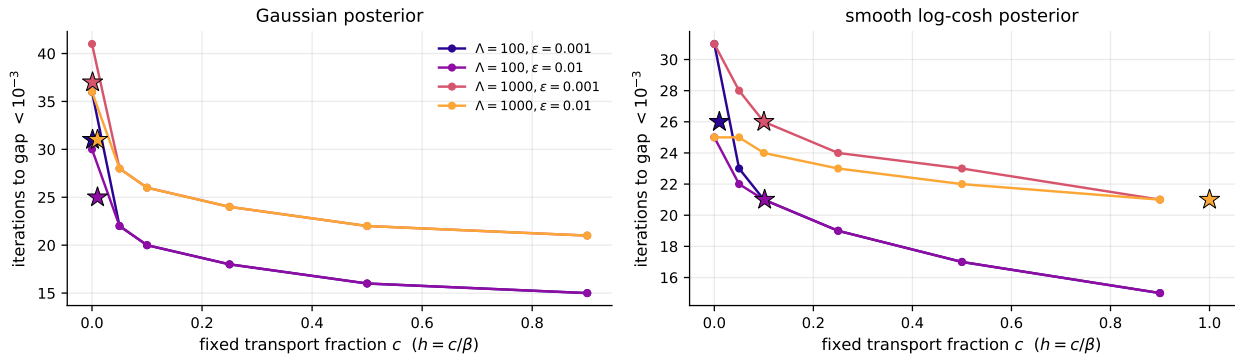


Figure 5: Schedule sweep: iterations to gap $< 10^{-3}$ vs the fixed transport fraction c ($h = c/\beta$), one curve per (Λ, ϵ) . $c = 0$ is FR-only; $\star$ marks the theory schedule $h = \mu_{\min}$.

Table 3: Per-run hitting iterations to reach energy gap below 10^{-1} , 10^{-3} , 10^{-6} , and the final gap, per target/ Λ/ε /method. ∞ denotes the threshold was not reached within the iteration budget.

target	method	Λ	ε	$n_{10^{-1}}$	$n_{10^{-3}}$	$n_{10^{-6}}$	gap _{final}
Gaussian	FR-only	100	0.001	30	36	44	0
	W-only	100	0.001	138	575	1264	0
	WFR-adaptive	100	0.001	9	15	23	0
	WFR-fixed	100	0.001	11	16	25	0
	WFR-theory	100	0.001	25	31	40	0
Gaussian	FR-only	100	0.01	24	30	39	0
	W-only	100	0.01	138	575	1264	0
	WFR-adaptive	100	0.01	9	15	23	0
	WFR-fixed	100	0.01	10	16	24	0
	WFR-theory	100	0.01	19	25	34	0
Gaussian	FR-only	1000	0.001	35	41	50	0
	W-only	1000	0.001	746	3653	∞	1.1×10^{-5}
	WFR-adaptive	1000	0.001	14	21	29	$< 10^{-12}$
	WFR-fixed	1000	0.001	15	22	31	$< 10^{-12}$
	WFR-theory	1000	0.001	30	37	45	0
Gaussian	FR-only	1000	0.01	29	36	44	0
	W-only	1000	0.01	746	3653	∞	1.1×10^{-5}
	WFR-adaptive	1000	0.01	14	21	29	$< 10^{-12}$
	WFR-fixed	1000	0.01	15	22	31	$< 10^{-12}$
	WFR-theory	1000	0.01	24	31	40	0
smooth log-cosh	FR-only	100	0.001	23	31	44	$< 10^{-12}$
	W-only	100	0.001	171	260	387	$< 10^{-12}$
	WFR-adaptive	100	0.001	9	16	29	$< 10^{-12}$
	WFR-fixed	100	0.001	10	17	29	$< 10^{-12}$
	WFR-theory	100	0.001	18	26	40	$< 10^{-12}$
smooth log-cosh	FR-only	100	0.01	17	25	39	$< 10^{-12}$
	W-only	100	0.01	171	260	387	$< 10^{-12}$
	WFR-adaptive	100	0.01	8	16	29	$< 10^{-12}$
	WFR-fixed	100	0.01	10	17	29	$< 10^{-12}$
	WFR-theory	100	0.01	13	21	34	$< 10^{-12}$
smooth log-cosh	FR-only	1000	0.001	23	31	44	$< 10^{-12}$
	W-only	1000	0.001	1680	2560	3813	$< 10^{-12}$
	WFR-adaptive	1000	0.001	14	22	35	$< 10^{-12}$
	WFR-fixed	1000	0.001	15	23	36	$< 10^{-12}$
	WFR-theory	1000	0.001	18	26	40	$< 10^{-12}$
smooth log-cosh	FR-only	1000	0.01	17	25	39	$< 10^{-12}$
	W-only	1000	0.01	1680	2560	3813	$< 10^{-12}$
	WFR-adaptive	1000	0.01	13	21	34	$< 10^{-12}$
	WFR-fixed	1000	0.01	14	22	35	$< 10^{-12}$
	WFR-theory	1000	0.01	13	21	34	$< 10^{-12}$

Table 4: Schedule sweep for WFR-fixed: fraction c , transport $h = c/\beta$, iterations to gap $< 10^{-3}$, and final gap.

target	(Λ, ε)	c	$h = c/\beta$	$n_{10^{-3}}$	gap _{final}
Gaussian	(100, 0.001)	0	0.0000	36	0
		0.05	0.0500	22	0
		0.1	0.1000	20	0
		0.25	0.2500	18	0
		0.5	0.5000	16	0
		0.9	0.9000	15	0
Gaussian	(100, 0.01)	0	0.0000	30	0
		0.05	0.0500	22	0
		0.1	0.1000	20	0
		0.25	0.2500	18	0
		0.5	0.5000	16	0
		0.9	0.9000	15	0
Gaussian	(1000, 0.001)	0	0.0000	41	0
		0.05	0.0500	28	0
		0.1	0.1000	26	0
		0.25	0.2500	24	0
		0.5	0.5000	22	$< 10^{-12}$
		0.9	0.9000	21	$< 10^{-12}$
Gaussian	(1000, 0.01)	0	0.0000	36	0
		0.05	0.0500	28	0
		0.1	0.1000	26	0
		0.25	0.2500	24	0
		0.5	0.5000	22	$< 10^{-12}$
		0.9	0.9000	21	$< 10^{-12}$
smooth log-cosh	(100, 0.001)	0	0.0000	31	$< 10^{-12}$
		0.05	0.0049	23	$< 10^{-12}$
		0.1	0.0098	21	$< 10^{-12}$
		0.25	0.0246	19	$< 10^{-12}$
		0.5	0.0492	17	$< 10^{-12}$
		0.9	0.0886	15	$< 10^{-12}$
smooth log-cosh	(100, 0.01)	0	0.0000	25	$< 10^{-12}$
		0.05	0.0049	22	$< 10^{-12}$
		0.1	0.0098	21	$< 10^{-12}$
		0.25	0.0246	19	$< 10^{-12}$
		0.5	0.0492	17	$< 10^{-12}$
		0.9	0.0886	15	$< 10^{-12}$
smooth log-cosh	(1000, 0.001)	0	0.0000	31	$< 10^{-12}$
		0.05	0.0005	28	$< 10^{-12}$
		0.1	0.0010	26	$< 10^{-12}$
		0.25	0.0025	24	$< 10^{-12}$
		0.5	0.0050	23	$< 10^{-12}$
		0.9	0.0090	21	$< 10^{-12}$
smooth log-cosh	(1000, 0.01)	0	0.0000	25	$< 10^{-12}$
		0.05	0.0005	25	$< 10^{-12}$
		0.1	0.0010	24	$< 10^{-12}$
		0.25	0.0025	23	$< 10^{-12}$
		0.5	0.0050	22	$< 10^{-12}$
		0.9	0.0090	21	$< 10^{-12}$

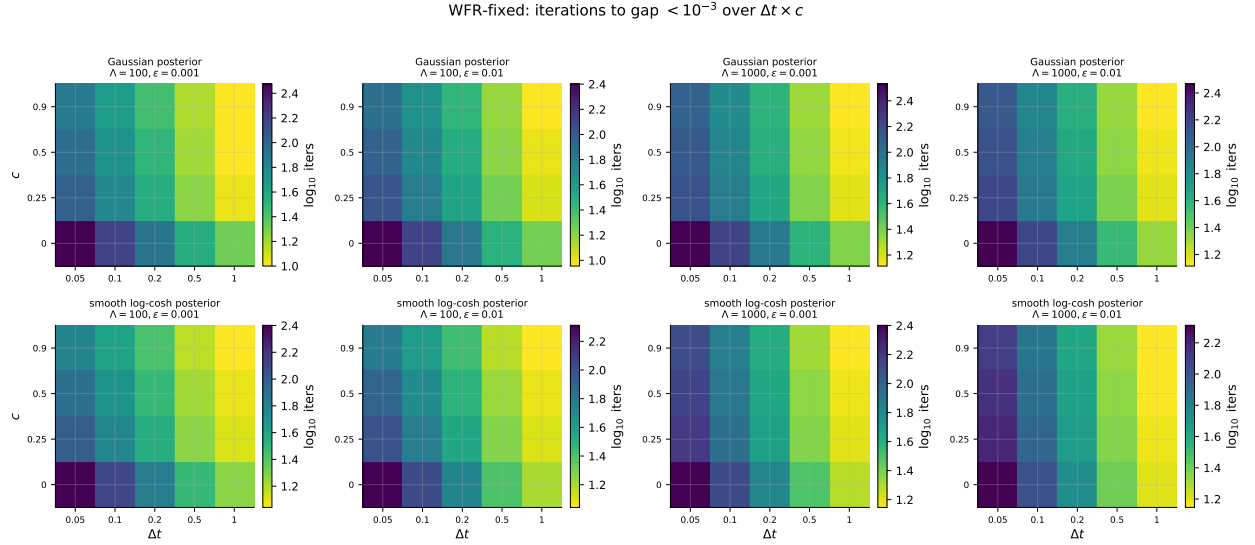


Figure 6: WFR-fixed: $\log_{10}$ iterations to gap $< 10^{-3}$ over the $\Delta t \times c$ grid, per target and (Λ, ϵ) . White cells were not SPD-feasible or did not reach the threshold within budget.